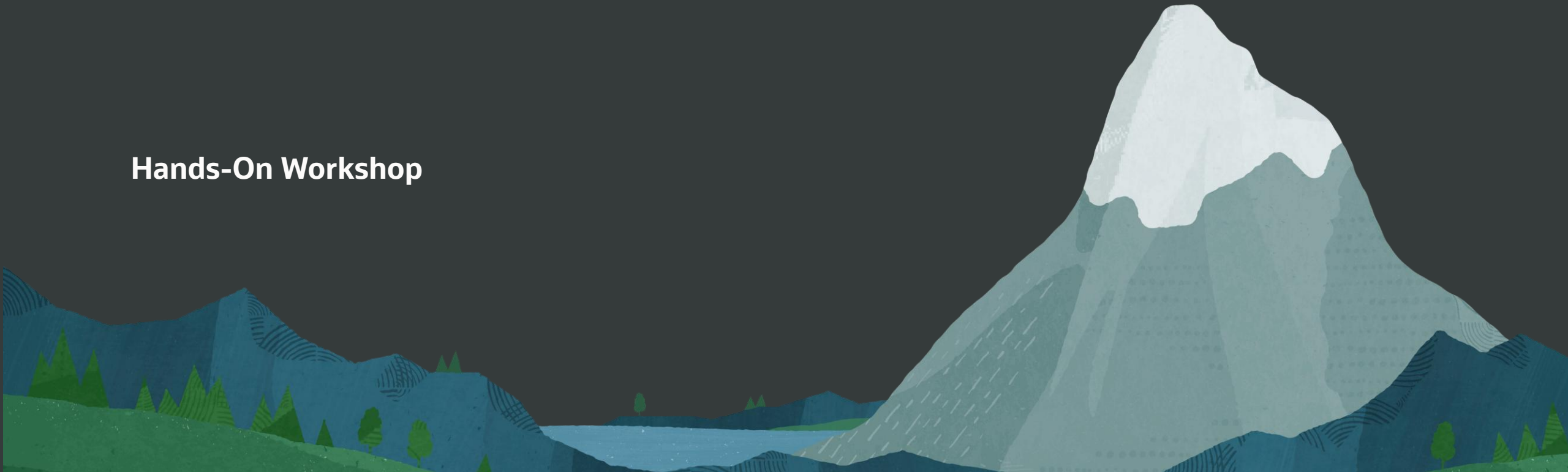


ORACLE | APEX

Oracle APEX

Build Scalable and Secure Low-Code Apps

Hands-On Workshop



Workshop Team



Arne Brüning



Rainer Willems



Workshop Agenda

Timing

09:00 – 12:30	Part One
12:30 – 13:30	Lunch break
13:30 – 17:00	Part Two



Just watch / Listen



Follow the Guide



We will see

1. Introduction
2. First Steps – First App
3. Application Adjustments
4. Layout – UI in APEX
5. Authentication & Authorization
6. Geographical Data
7. RESTful Services
8. Reports
9. Calendar
10. JavaScript in APEX
11. Charts

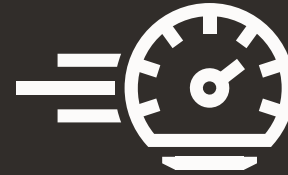
12. Working with Files
13. Data Loading
14. Mobile Applications
15. Approvals & Workflows
16. Individual Form Handling
17. Translations
18. Application Lifecycle
19. AI in APEX
20. Printing/PDF
21. Miscellaneous
22. Links & More

What & Why

Low-Code application platform (**LCAP**) is used to rapidly develop and deploy custom applications by abstracting and minimizing hand coding.

At a minimum, an LCAP must include low-code capabilities (such as a model-driven or graphical programming approach with scripting) to develop a complete application consisting of user interfaces, business logic, workflow, and data services.

- Gartner



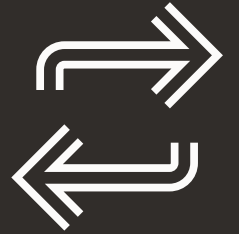
Speed

Easy to learn
& Faster
development



Innovation

More time to
focus on
making
better apps



Agility

Respond to
user requests
more quickly

What is Oracle APEX?

Oracle's Strategic Low-Code Application Platform

APEX is the **world's most popular** enterprise **low-code** application platform.

It enables you to build scalable and secure enterprise apps with world-class features **20X** faster with **100X** less code.

Oracle APEX delivers the most productive way to develop and deploy **Mobile, Web, and Desktop** apps everywhere - cloud and on-premises.



ā'pěks
(#orclapex)

21 Million +

apps created till date

7,000+

APEX apps created daily

850,000+

APEX developers worldwide

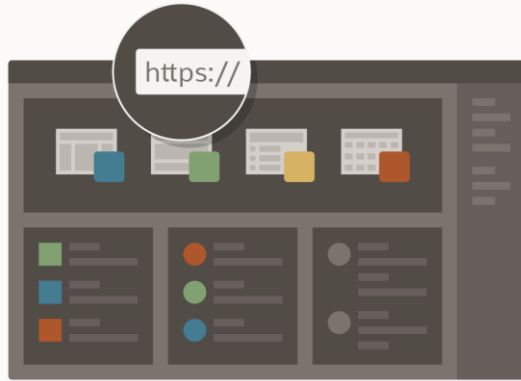
100,000+

customers across 150 countries

90%

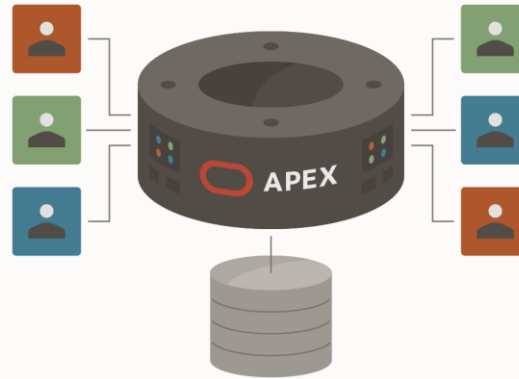
of Fortune 500 use Oracle APEX

Distinguishing Characteristics



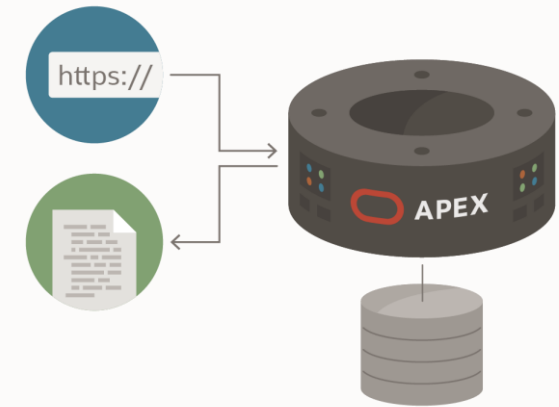
App Development IDE is a web browser

No client software needed



App Definitions are stored in the database as meta data

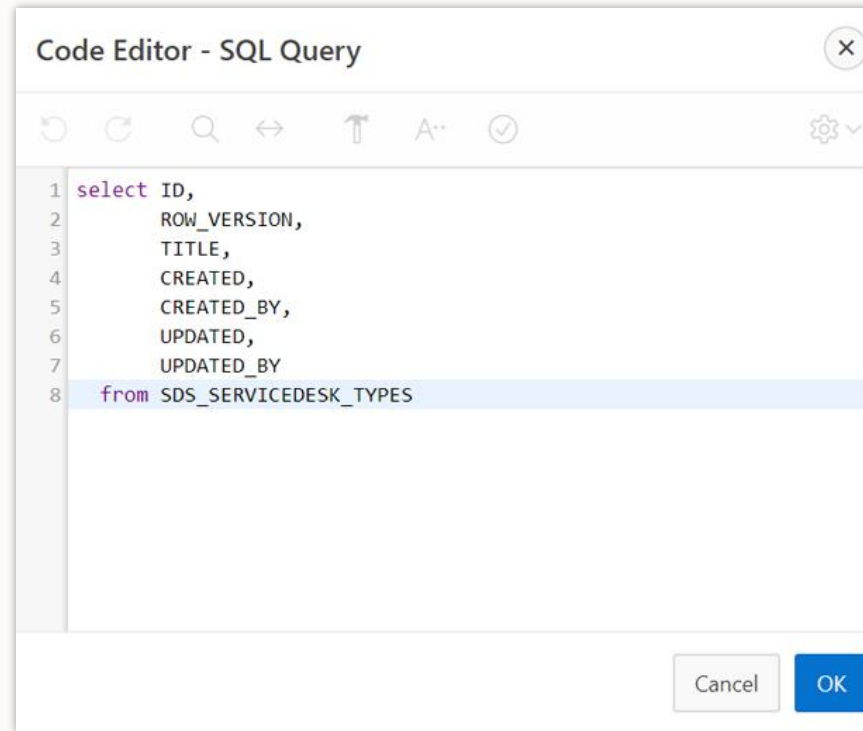
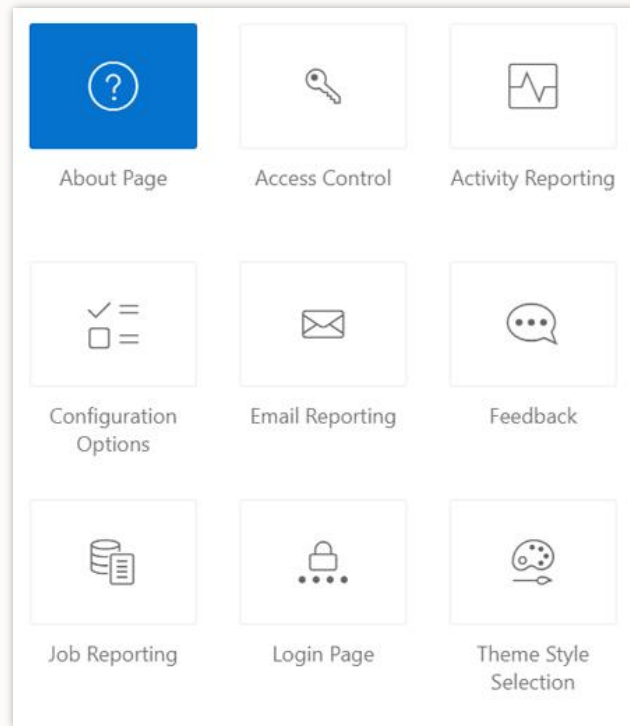
Declarative – No code generation



Page generation is efficient with only one request and one response

Data processing done in the Database

Oracle APEX – Unconstrained!



```
import React, { useState } from "react";
import falconForm, { isRequired, isEmail } from "falcon-form";

var renderCount = 0;

const App = () => {
  let [formInitialValues] = useState({
    email: "",
    name: ""
  }); // getInitialValues, use setFormInitialValues to set data from API

  const formSubmitAction = values => {
    console.log("Form submitted", values);
  }; // declare submit success action

  const fieldValidators = {
    email: [isRequired, isEmail],
    name: [isRequired]
  }; // declare field validators

  const { values, errors, fieldChange, formSubmit } = falconForm(
    formInitialValues,
    formSubmitAction,
    fieldValidators
  );
  // pass initial values, submit actions, field validators
  // use values, errors and field handlers in form

  renderCount += 1;
  return (
    <div>
```

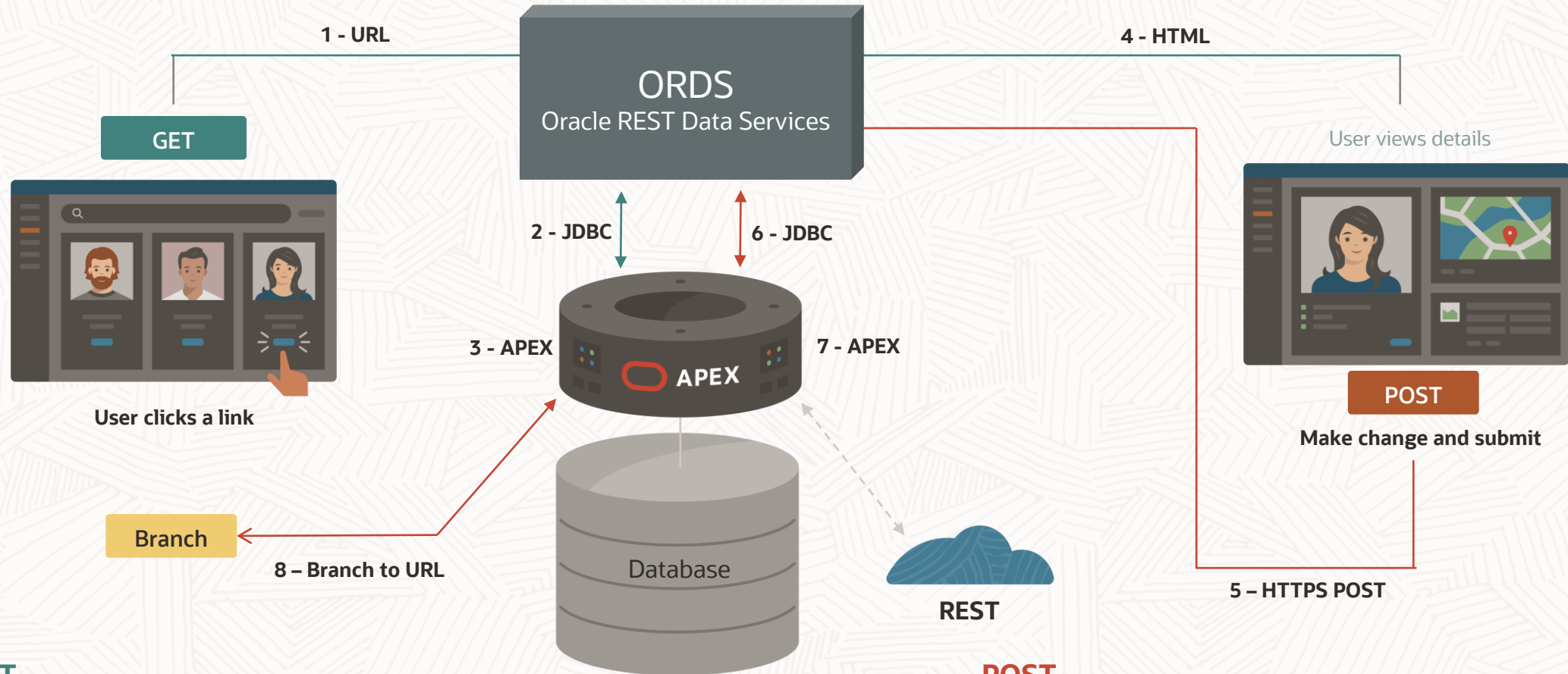
Oracle APEX uniquely spans the App Dev spectrum, from No Code to Low Code to High Control

User Interface for Any Device



**Modern, intuitive, accessible UI for any device
fully customizable and responsive apps**

APEX Architecture



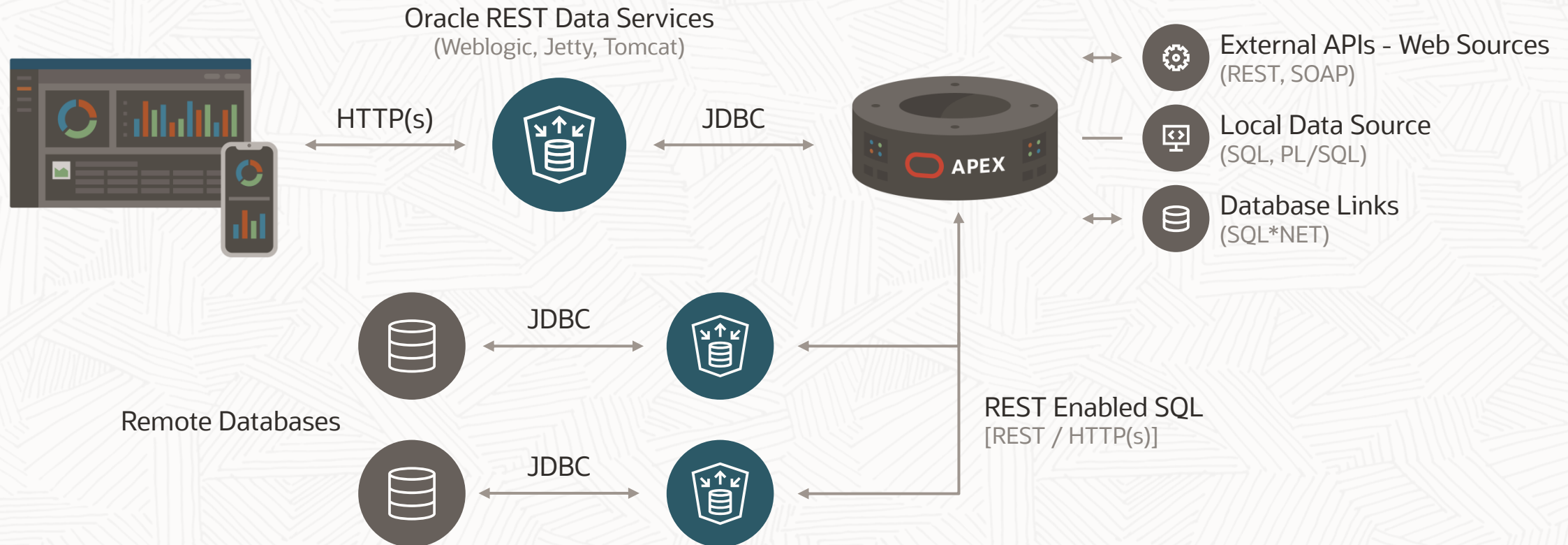
GET

1. User clicks a link which sends HTTPS GET request to ORDS
2. ORDS makes a database JDBC call to the APEX Engine
3. APEX generates HTML page (full page or partial)
4. ORDS returns the generated HTML back to the browser

POST

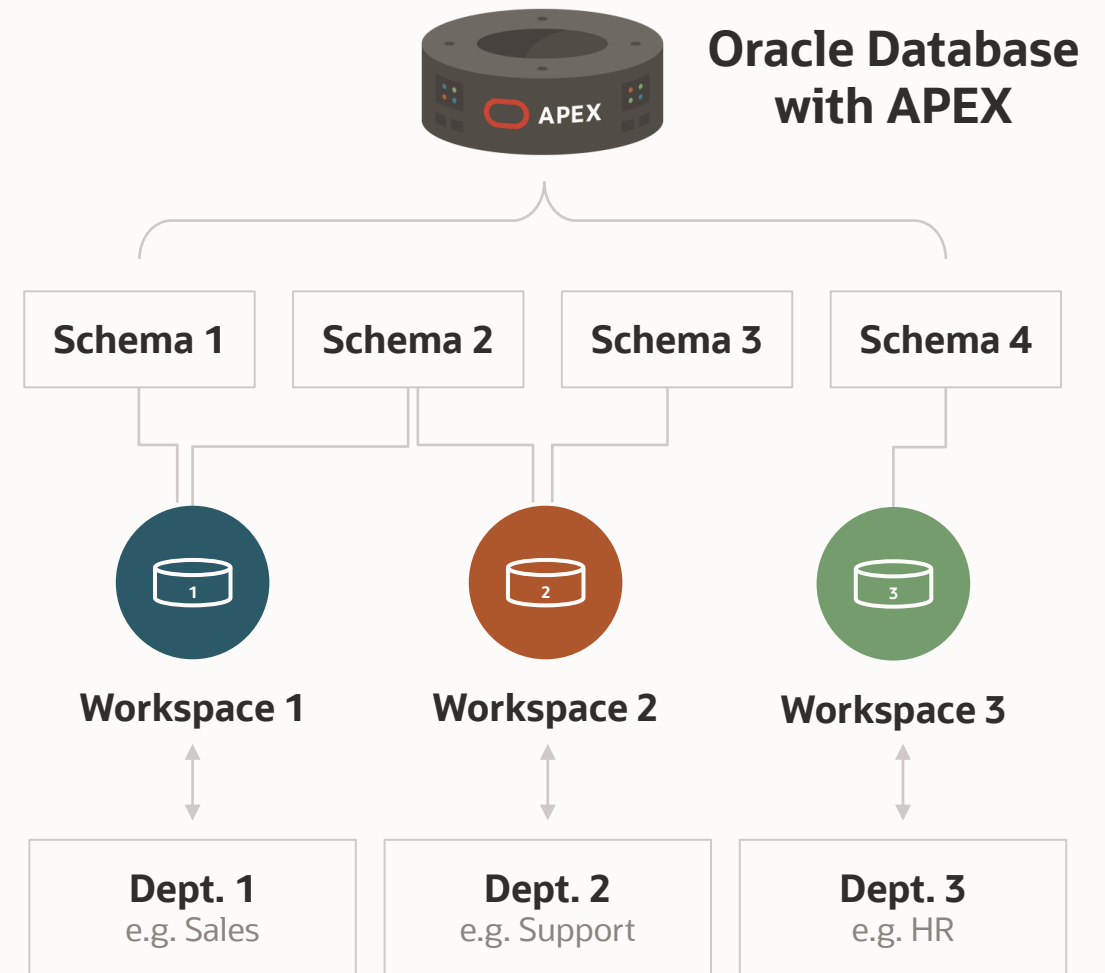
5. User clicks a button which does an HTTP POST to ORDS
6. ORDS makes a database JDBC call to the APEX Engine
7. APEX processes the request
8. APEX determines URL and branches

APEX Data Sources

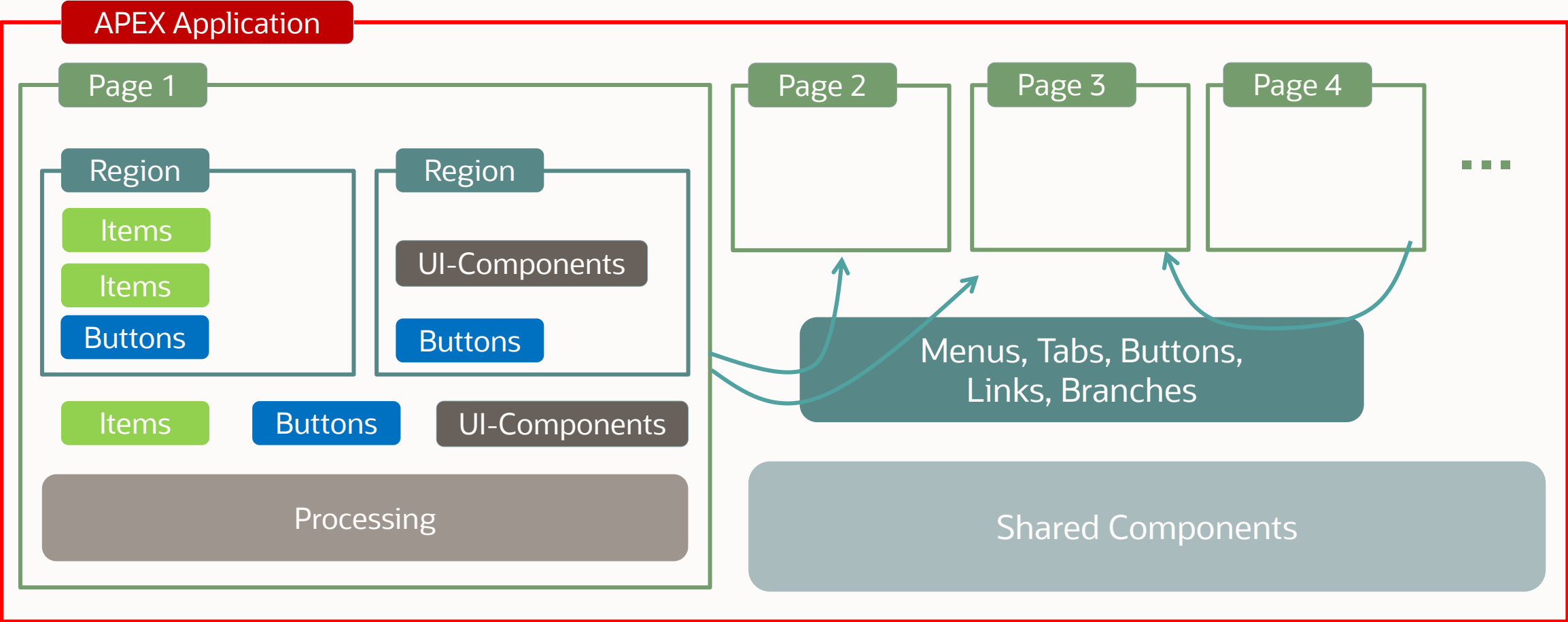


Single Database Instance and Multiple Workspaces

- Workspaces used to define application definitions / Schemas hold data
- The many-to-many relationship between Workspaces and Schemas
- Instance Administrators manage the environment and schema access
- Departments can request more space and access to a new schema
- For example, Oracle's internal-only service <https://apex.oraclecorp.com/en/> has over 9000 Workspaces, covering every line of business in Oracle



Structure of an APEX Application



Region & Item Types

Item-Types

Checkbox	Password
Checkbox Group	Percent Graph
Color Picker	Popup LOV
Combobox	QR Code
Date Picker	Radio Group
Display Image	Rich Text Editor
Display Map	Select List
Display Only	Select Many
File Upload	Select One
Geocoded Address	Shuttle
Hidden	Star Rating
Image Upload	Switch
List Manager	Text Field
Markdown Editor	Text Field with autocomplete
Number Field	Textarea

Region-Types

Breadcrumb
Calendar
Cards
Chart
Classic Report
Dynamic Content
Faceted Search
Form
Help Text
Interactive Grid
Interactive Report
List
Map
Region Display Selector
Search
Smart Filters
Static Content
Tree
URL
Workflow Diagram
+
Theme Components

APEX

App Builder

SQL Workshop

Team Development

Gallery

Application 154280

Page Designer

3 Panes

Toolbar

Page 4: EmpFacets

Pre-Rendering

Components

Breadcrumb Bar

Left Column

Dynamic Actions

Processing

Shared Components

Rendering

P4_SEARCH

P4_DEPTNO

P4_MGR

P4_JOB

P4_SAL

Body

Button Bar

Next

RESET

Search Results

Actions

Sort Order

P4_ORDER_BY

Post-Rendering

Layout

Page Search

Help

Search

COPY

EDIT

PREVIOUS

NEXT

SORT ORDER

REGION BODY

REGION CONTENT

SUB REGIONS

CLOSE

HELP

DELETE

CHANGE

CREATE

BODY

Button Bar

PREVIOUS

CLOSE

DELETE

REGION BODY

REGION CONTENT

CHANGE

SUB REGIONS

EDIT

CREATE

NEXT

RESET

Search Results

SORT ORDER

P4 ORDER BY

Regions

Items

Buttons

Checkbox

Checkbox Group

Color Picker

Combobox

Date Picker

Display Image

Display Map

Display Only

File Upload

Geocoded Address

Hidden

Image Upload

List Manager

Markdown Editor

Number Field

Region

Attributes

Filter

Identification

Title

Type

Faceted Search

Source

Filtered Region

Search Results

Layout

Sequence

10

Parent Region

No Parent

Position

Left Column

Appearance

Template

Standard

Template Options

Use Template Defaults, Remove Body Padding, Hidden but accessible, Scroll - Default

CSS Classes

Icon

Render Components

Above Content

Accessibility

Use Landmark

Landmark Type

Template Default

3 Panes

Toolbar

Shared Components

Processing

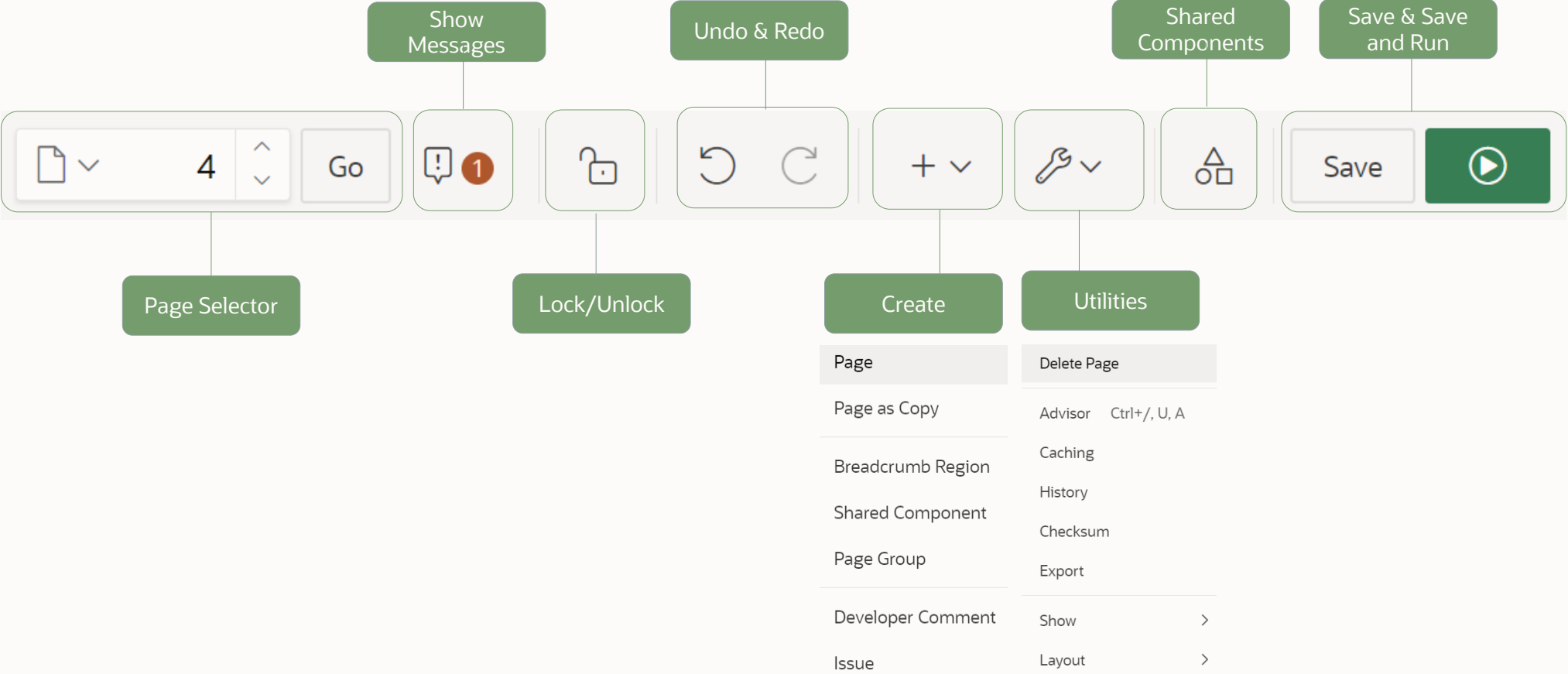
Dynamic Actions

Rendering

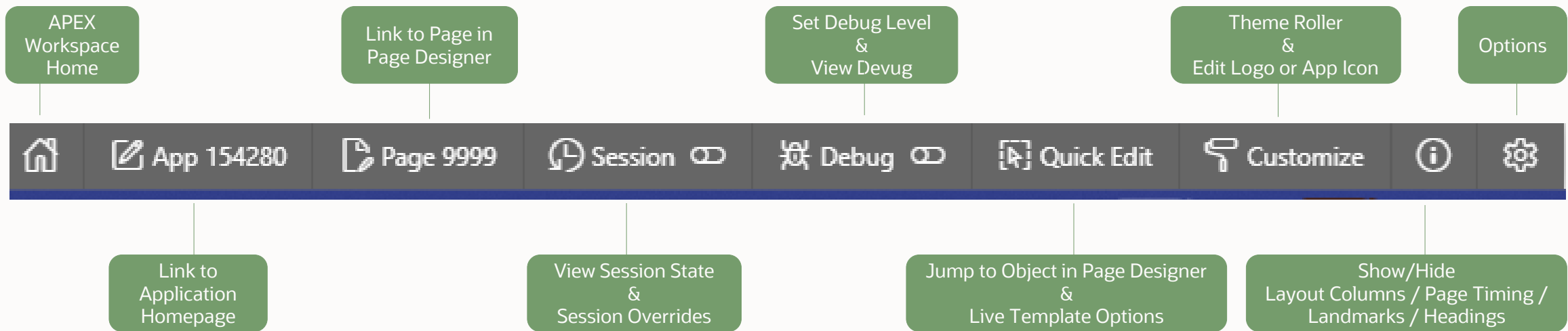
Type dependent properties

Properties

Toolbar



Developer Toolbar



Our mission is to help people see
data in new ways, discover insights,
unlock endless possibilities.

